HAXLR8 3.0

NATIONAL LEVEL HACKATHON • MIT MYSORE

HEALTHCARE PROBLEM STATEMENTS

15 Challenges | Pick one. Build it. Save lives.

Practical healthcare innovation through functional prototypes, technical problem-solving and multidisciplinary collaboration.

1

Real-Time ICU Patient Monitoring and Intelligent Alert Prioritization

ICU patients require continuous monitoring of vital parameters, generating multiple alerts that may overwhelm healthcare professionals. Delayed identification of critical conditions can lead to serious patient complications. Design an intelligent patient monitoring system that continuously tracks vital parameters, identifies abnormal physiological patterns, prioritizes critical alerts, and ensures timely notifications to healthcare staff.

2

Continuous Cold-Chain Assurance for Vaccines and Blood Supplies

Vaccines and blood products transported across hospitals, healthcare centres, and remote locations are frequently exposed to temperature fluctuations that can compromise their quality and safety. Design an intelligent cold-chain monitoring solution that continuously tracks environmental conditions, detects temperature violations, records transportation history, and generates immediate alerts whenever storage conditions exceed permissible limits.

3

Automated Hand-Hygiene Compliance Measurement for Hospital-Acquired Infection Prevention

Hospital-acquired infections remain a significant healthcare concern, while hand-hygiene compliance is often monitored through manual observations and periodic audits. Design an automated monitoring system that detects hand-hygiene activities, identifies compliance gaps, and generates actionable reports to help healthcare facilities improve infection prevention practices.

4

Automated Asset Tracking and Utilization Optimization of Critical Medical Equipment

Hospitals frequently experience difficulties locating essential movable medical equipment such as wheelchairs, infusion pumps, patient monitors, and emergency devices. This results in equipment underutilization, operational delays, and unnecessary procurement. Design an intelligent asset tracking solution that provides real-time equipment visibility, monitors utilization, and supports efficient allocation across hospital departments.

5

Wearable-Based Post-Discharge Recovery Monitoring and Rehabilitation Tracking

Patients discharged after intensive care or major medical treatment often experience physical and functional difficulties during their recovery period. However, continuous monitoring and rehabilitation follow-up are limited once patients leave the hospital. Design a wearable-based monitoring solution that tracks relevant health indicators and physical activity, identifies potential recovery concerns, and supports continuous rehabilitation monitoring through timely alerts.

6

Smart Village-Level Health Screening and Real-Time Abnormality Detection

Rural communities frequently lack access to continuous healthcare screening and timely identification of abnormal health conditions. Existing screening activities often depend on manual measurements and delayed reporting. Design a portable smart healthcare screening solution that measures essential physiological parameters, identifies abnormal readings, stores screening records, and enables timely alerts and referrals to healthcare professionals.

7

Intelligent Emergency Response and Ambulance Patient Monitoring System

Emergency patients often lose valuable treatment time due to traffic delays, inappropriate hospital selection, and insufficient communication between ambulances and receiving hospitals. Design an intelligent emergency response system that monitors patient conditions during transportation, supports location-based ambulance tracking, identifies suitable healthcare facilities, and enables real-time information sharing with hospitals before patient arrival.

8

Real-Time Critical Care Bed and Resource Availability Monitoring

During emergencies, disasters, and mass-casualty situations, hospitals face difficulties maintaining accurate information about available critical care beds, ventilators, and essential medical resources. This can delay patient admission and emergency treatment. Design an automated resource monitoring solution that tracks critical care resource availability, updates occupancy information in real time, and provides centralized visibility to support faster patient allocation.

9

Smart IV Infusion Monitoring and Abnormal Flow Detection System

Intravenous fluid administration requires continuous supervision to ensure that the prescribed infusion is delivered safely and without interruptions. Empty fluid bags, abnormal flow rates, blockages, and unexpected interruptions can create risks for patients. Design an intelligent IV infusion monitoring system that tracks fluid levels and flow rates, identifies abnormal infusion conditions, and generates immediate alerts for healthcare staff.

10

Portable Smart Respiratory Monitoring and Early Distress Detection System

Respiratory abnormalities can develop rapidly in patients suffering from respiratory illnesses, post-operative complications, or other critical conditions. Conventional monitoring may not always provide continuous visibility into breathing patterns outside intensive care environments. Design a portable respiratory monitoring solution that continuously measures breathing rate and respiratory patterns, identifies abnormal breathing conditions, and generates timely alerts to caregivers for early intervention.

11

Predictive Medication Safety Verification for Elderly Patients on Complex Multi-Drug Regimens

Elderly patients are frequently prescribed multiple medications for different health conditions, increasing the risk of harmful drug interactions, duplicate prescriptions, incorrect dosages, and medication-related complications. Design an intelligent medication verification platform that analyses prescriptions against patient medication histories, identifies potential medication risks, and provides timely recommendations to support safer prescribing decisions.

12

AI-Based Specialty Matching for Long-Term Undiagnosed Patients

Many patients experience prolonged diagnostic delays because their symptoms and medical histories are not effectively connected to the appropriate medical specialties. Repeated consultations and fragmented patient records can further delay appropriate treatment. Design an AI-powered patient assessment solution that analyses reported symptoms, medical history, and available clinical information to recommend relevant medical specialties and support timely clinical evaluation.

13

Predictive Medicine Stock-Out and Intelligent Supply Replenishment

Rural healthcare centres and primary health facilities frequently experience shortages of essential medicines due to unpredictable consumption, delayed replenishment, and inefficient inventory management. These shortages can interrupt treatment and force patients to seek medicines elsewhere. Design a predictive inventory management solution that analyses medicine consumption patterns, forecasts potential stock-outs, identifies replenishment requirements, and supports timely medicine supply planning.

14

Intelligent Clinical Handoff and Guaranteed Information Continuity

Critical patient information can be missed or incorrectly communicated when responsibility is transferred between successive hospital care teams during shift changes. Incomplete handovers can lead to medication errors, delayed treatments, and gaps in patient care. Design an intelligent clinical handoff platform that organizes patient information, tracks ongoing treatments and pending tasks, identifies missing handover details, and ensures structured information continuity between healthcare teams.

15

Intelligent Phone-Based Medicine Ordering and Replenishment for Rural Households

Rural households without reliable internet connectivity or access to smartphones often face difficulties checking medicine availability and arranging prescription refills. Patients may need to travel long distances to healthcare facilities only to discover that the required medicines are unavailable. Design an intelligent phone-based medicine ordering solution that enables patients to check medicine availability, place verified medicine requests, track order status, and coordinate medicine delivery through voice-based or telephone services without requiring internet access.

Common Participant Guidelines

- Participants should develop a functional prototype that addresses the selected healthcare challenge.
- Hardware-oriented solutions should demonstrate meaningful physical hardware integration wherever applicable.
- Software-oriented solutions should demonstrate a working application, platform or AI-based implementation.
- Teams may use suitable development boards, sensors, open-source tools, APIs and publicly available datasets.
- Solutions will be expected to demonstrate innovation, technical feasibility, scalability and practical healthcare impact.
- Any AI-generated medical recommendation must be presented as decision support and not as a replacement for qualified medical professionals.

Select your problem statement and register for HAXLR8 3.0

